

Chapter 9 Warm-Up: Organizing Course Data with Lists

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1 Nested Course Lists

- Chapter 9 develops techniques for modifying, sorting, and searching lists. We will use a nested list to represent a college course schedule. One course is represented by

```
course = ["PHYS", 100, "SLC", 202]
```

The four elements represent

```
[subject, number, building, room]
```

Thus, this record represents Physics 100 in SLC 202.

2 A List of Lists

- Several course records can be stored inside one master list:

```
courses = [  
    ["PHYS", 100, "SLC", 202],  
    ["MATH", 150, "SLC", 105],  
    ["BIOL", 101, "SLC", 210],  
    ["PHYS", 200, "SLC", 202]  
]
```

Each element of `courses` is itself a list. Print one complete record:

```
print(courses[1])
```

Then print only its room number:

```
print(courses[1][3])
```

3 Appending Courses

- Define a function that adds a course to the master list:

```
def add_course(courses, course):  
    courses.append(course)
```

Test it using

```
new_course = ["COSC", 120, "SLC", 416]  
add_course(courses, new_course)
```

Use `print(len(courses))` to verify that the new record was added.

4 Sorting the Master List

- Lists can be sorted using `sort()`. Define

```
def sort_courses(courses):  
    courses.sort()
```

Then run

```
sort_courses(courses)  
for course in courses:  
    print(course)
```

Since the subject is element 0, Python first sorts the records alphabetically by subject. If two subjects are the same, which element is compared next?

5 Searching for Room Conflicts

- Two courses conflict if they use the same building and room. Define

```
def same_room(c1, c2):  
    return c1[2] == c2[2] and c1[3] == c2[3]
```

Now use nested loops to compare every pair:

```
for i in range(len(courses)):  
    for j in range(i+1, len(courses)):  
        if same_room(courses[i], courses[j]):  
            print("Room conflict:")  
            print(courses[i])  
            print(courses[j])
```

Why does the inner loop begin at `i+1` instead of zero?

6 Warm-Up Exercise

- Add the following records using `add_course()`:

```
["MATH", 250, "SLC", 105]  
["CHEM", 110, "SLC", 210]  
["PHYS", 150, "SLC", 310]
```

Complete the following:

- Sort the master list
 - Print every course with a `for` loop
 - Use the nested-loop search to print every room conflict
- Before running each bit of code, predict which new courses will cause room conflicts.